

Chapter 3 Review Assignment

Overview

In this assignment, you will review the material covered in chapter 3 of Automate the Boring Stuff.

Objectives

- Review the use of `while` loops in Python.
- Review the use of `try except` statements for error handling.
- Review the use of user-defined functions.
- Enhance understanding of error handling in Python
- Practice implementing input validation to prevent errors.
- Promote good programming practices through the use of user prompts, robust error checking, and providing clear feedback when inputs are invalid.

Requirements

User Input

Your program should begin by calling the `get_input` function to retrieve a valid input from the user.

Program Output

Based on the user's input, your program should either notify the user that they have not provided a valid number and ask for valid input, or print out the square of the number they provided.

- Example: if the value returned from `get_input` is 5.0 and the result of TODO Step 6 is 25.0
- The value printed would be: `The square of 5.0 is 25.0.`

Global Scope TODOs

In the global scope you have three TODO tasks that must be complete. These TODO tasks involve using the `get_input` function, raising the value returned from `get_input` to the power of two, and printing the result to the screen.

Functions to Implement

1. `get_input`

- Complete the TODO tasks in `get_input` to ensure that only a valid number is provided by the user. If the user provides an invalid number such as "twenty", you should output an error message that indicates they need to provide a valid number. Design your error message as you see fit, the auto-grader will not deduct based on the formatting or contents of your error message.